

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Continuous Assignment 02

Submission:

Students are required to design, simulate, monitor, and document the following IoT-based systems using **Wokwi**, **VS Code**, **MQTT Broker**, and **Node-RED Dashboard**.

For each question, take clear screenshots of:

- Circuit design in Wokwi
- Complete source code (copy & paste)
- Serial Monitor output in VS Code (showing multiple readings)
- Node-RED Dashboard showing live data and history

Organize the PDF with clear section headings (e.g., “Question 1 – Circuit”, “Question 1 – Code”, “Question 1 – Serial Monitor”, “Question 1 – Node-RED Dashboard”, etc.). Combine everything into **one single PDF** file and submit.

Questions 01: Smart Water Tank Monitoring and Control System

The Smart Water Tank Monitoring System continuously monitors the water level and controls a water pump automatically. Sensor readings are published to an MQTT broker and visualized on a Node-RED dashboard for remote supervision.

Hardware Components

- ESP32 Development Board
- Ultrasonic Sensor (HC-SR04)
- Relay Module or LED (to simulate the water pump)

System Behavior

1. Water Level Detection

- a. Ultrasonic sensor measures distance every 2 seconds.
- b. Assume tank height = 200 cm.
- c. Water level percentage = $((200 - \text{distance}) / 200) \times 100$ (clamp between 0% and 100%).
- d. Display the distance (cm) and water level (%) on the Serial Monitor.

2. Automatic Pump Control

- a. Water level $< 30\%$
 - i. Pump turns ON
 - ii. MQTT message published: "Pump ON – Low Water Level"
- b. Water level between 30% and 80%
 - i. Pump remains OFF
 - ii. MQTT message published: "Normal Level"
- c. Water level $> 80\%$
 - i. Pump remains OFF
 - ii. MQTT message published: "Tank Full"

MQTT Topics

- home/tank/level – Water level percentage (integer)
- home/tank/pump – Pump status ("ON" or "OFF")

Node-RED Dashboard Requirements

- Gauge widget: Water level percentage (0–100)
- Text widget: Current pump status
- Chart widget: Water level history over time
- Button widget: Manual override (Pump ON/OFF)

Question 02: Smart Energy Monitoring and Load Control System

The Smart Energy Monitoring System simulates energy consumption monitoring and allows remote control of electrical loads. The system publishes energy data via MQTT and provides real-time visualization and control through Node-RED.

Hardware Components

- ESP32 Development Board
- Potentiometer (to simulate power consumption)
- Two LEDs (to simulate electrical loads – Load 1 and Load 2)

System Behaviour

1. Energy Monitoring

- a. Read potentiometer value every 1 second.
- b. Map the analog value (0–4095) to power percentage 0–100%.
- c. Display raw value and power percentage on the Serial Monitor.
- d. Publish power value to MQTT.

2. Load Control Logic

- a. Power usage $> 65\%$
 - i. Load 1 turns OFF automatically
 - ii. Publish alert message: "High Power – Load 1 OFF"
- b. Power usage $\leq 65\%$
 - i. Both loads operate normally (can be controlled remotely).

MQTT Topics

- home/energy/power – Power consumption percentage (0–100)
- home/energy/load1 – Load 1 status ("ON" / "OFF")
- home/energy/load2 – Load 2 status ("ON" / "OFF")
- home/energy/alert – Alert messages (when Load 1 is auto-turned off)

Node-RED Dashboard Requirements

- Gauge widget: Current power consumption (%)
- Chart widget: Power usage history
- Toggle switches: Remote control for Load 1 and Load 2
- Text widget: Display latest energy alert messages